[INCH-POUND]
OO-S-2896
March 14, 1997
SUPERSEDING
MIL-S-29208B(YD)
17 January 1992

FEDERAL SPECIFICATION

SPREADER, MATERIAL, POWER DRIVEN, TRUCK MOUNTED, COMMERCIAL

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a hydraulically driven material spreader for spreading sand and aggregates or abrasive materials on road surfaces.

1.2 Classification. Spreaders shall be of the following types, as specified (see 6.2).

Type I - Truck Chassis Mounted.
Type II - Dump Body Mounted.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 1581), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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Federal Standards

FED-STD-595	-	Colors Used in Government Procurement.
FED-STD-794	-	Trucks and Truck Tractors, Medium Commercial, 4 X 2 and 4 X 4, 7,250 to 16,000 KG (16,000 to 35,000 lbs) GVW.

Military Standard

MIL-STD-209	-	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of federal and military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of the invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The spreader shall consist of an all steel hopper type body complete with auxiliary engine or hydraulic power driven auger or conveyor, rear-end material spreader and hydraulic system. The spreader shall be complete with all components and accessories to provide a complete and serviceable unit.

3.2 First article. When specified (see 6.2), a sample shall be subjected for first article inspection in accordance with 4.2.1.

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3.3 Standard commercial product. The spreader shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the spreader being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Safety. All rotating or reciprocating parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded.

3.4.1 Noise level. Operation of the spreader shall not cause the vehicle operators to experience noise levels greater than or equal to 85 dB(A). If the spreader does cause operator's noise levels to reach or exceed 85 dB(A), then hearing protection warnings will be placed on the spreader where they can be readily seen by the operators.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5.1 Prohibited materials. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by Title 10, Code of Federal Regulations (CFR), Parts 30 and 40, and other radioactive material in which the specific gravity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.6 Construction. The equipment shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.7 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.8 Components.

3.8.1 Hopper body. The hopper body shall have a water level capacity of not less than 9 cubic yards (6.9 cubic metre). The inside length of the hopper shall be not less than 16 feet

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(4 877 millimetre (mm)). The inside width of the hopper shall be not greater than 80 inches (2 032 mm). The hopper body shall be all-welded construction of not less than 10 gauge (3.4 mm) stainless steel. Body shall have not less than four upright side supports on each side of the body. Body support longitudinals shall be not less than 10 gauge (3.4 mm) stainless steel. Body shall have sloping sides to prevent bridging of material. The rear discharge gate shall be adjustable.

3.8.2 Grid. A stainless steel grid shall be provided on top of the body. Grid shall be constructed of stainless steel rods not less than 0.25-inch (6 mm) in diameter. The rods shall be spaced not less than 2.25 inches (57 mm) apart on centers in one direction and not less than 1.5-inch (38 mm) apart on centers in the other direction. The grid shall be constructed of four center hinged sections for easy handling. Stainless steel hinge rods shall be not less than 0.50-inch (13 mm) in diameter and shall run continuously from front to rear of the body.

3.8.3 Conveying system. An auger or chain conveyor shall be used to convey the material from the hopper body to the rear discharge chute.

3.8.3.1 Auger. The steel auger shall have a progressive pitch flighting approximately 7-inch (178 mm) outside diameter (od), welded around a 4-inch (102 mm) od tube. Auger flighting shall be not less than 0.50-inch (13 mm) thick at the base and 0.375-inch (10 mm) thick at the outer edge. The outer periphery of the flights shall be mounted on self-aligning bearings on the outside of the bunker ends for ease in disassembly. The auger trough shall be easily replaceable and constructed of not less than 10 gauge (3.4 mm) steel.

3.8.3.2 Conveyor and conveyor chain. The conveyor shall be constructed of 10 gauge (3.4 mm) hi-grade carbon steel. The conveyor floor shall have a flat design of replaceable abrasion resistant steel. Conveyor shafts shall have dust-sealed, heavy-duty, self-aligning ball bearings equipped with greased fittings. The conveyor chain shall be of the self-cleaning pintle type, manufactured of formed alloy steel with heat treated hardened pins and chain strands of not less than 21,000 pounds (144 795 kilopascals) tensile strength. The chain link strands shall be protected in a manner which will only expose the drag bar to the material. An adjustment device shall be provided to keep the chain tight.

3.8.4 Power system. The auger or conveyor and spinner shall be powered by independent cab-controlled hydraulic motors. A truck-mounted power takeoff or auxiliary gasoline engine shall be used to power a gear or vane type hydraulic pump. The hydraulic power system shall be complete including reservoir, valves, filters, and quick-disconnect couplers. Hydraulic hose shall conform to SAE J517. All operation controls shall be for cab mounting. The auxiliary gasoline engine, when furnished, shall be a four-cycle, air-cooled engine of not less than 18 horsepower (13 kilowatt), with electric start, charging system, ammeter, oil pressure gage (if required), and a hood for weather protection. Operating controls shall be mounted in the truck cab.

3.8.5 Spinner and spinner housing. The spinner and spinner housing shall be either detachable type or hinged type which raises and attaches to rear hopper body. The spinner shall be not less than 18 inches (457 mm) in diameter and the thickness of the replaceable fins (blades) shall be not

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less than 0.125-inch (3 mm). The spinner shaft, bearings, and gears shall be fully enclosed and lubricated. The spinner shall be powered hydraulically. The spinner assembly shall have not less than 6 inches (152 mm) of vertical adjustment. The spinner housing shall be fabricated of not less than 12 gauge (2.7 mm) steel, and shall have an internal deflector for directing material to discharge gate. The spinner housing shall be equipped with not less than three external deflectors.

3.8.6 Mounting.

3.8.6.1 Type I. The spreader shall be mounted on a 4 by 2, 28,000 pound (12 712 kilograms (kg)) minimum gross vehicle weight (GVW), 84-inch (2 134 mm) cab-to-axle dimension, diesel-engine-driven truck chassis conforming to FED-STD-794. Mounting of the spreader on the truck chassis shall be manufacturer's standard practice as furnished to the commercial market.

3.8.6.2 Type II. The spreader shall be designed for mounting in a standard steel dump body of a 34,500 pound (15 436 kg) GVW, 6X4 truck chassis. The manufacturer's standard holddown devices for mounting the spreader unit in the dump body shall be provided.

3.9 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubrication points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound per square inch (6 894 kilopascals) force or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with the type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.10 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag, and spatter, loose mill scale, water, dirt, corrosion product, or any other interfering substances. As soon as practicable, after cleaning, and before any coating interfering material can result, the surfaces shall be prepared to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils (0.063 5 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall be manufacturer's standard yellow, nearest match to No. 13538 yellow, conforming to FED-STD-595. The end item, allied equipment, and attachments shall be the same color.

3.11 Lifting and tiedown attachments. When specified (see 6.2), the spreader shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically

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attached to the spreader. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the spreader on the carrier when shipped.

3.12 Identification plate. An identification plate will be furnished by the contracting officer for each spreader. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each spreader in a conspicuous place with nonferrous screws, rivets, or bolts not less than 0.125-inch (3 mm) in diameter. The applicable nomenclature contained in the current contract item description shall be placed in the top blank.

3.13 Instruction plates. The spreader shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, rivets, or bolts of not less than 0.125-inch (3 mm) in diameter.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.14.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.14.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting’s ability to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable reference documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one spreader when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. When a first article test is specified by this contract and the contractor desires to deliver the test unit as a contract item, it shall be delivered as the last item on the contract only after the contractor, at that contractor's cost and expense, shall have completely cleaned, reconditioned, and overhauled the unit, making such replacements and modifications thereto as are required to make the unit acceptable as a contract item.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3 and the operation test of 4.4.3.

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4.3 Examination. Each spreader shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Failure to pass any test shall be cause for rejection.

4.4.1 Capacity. Capacity of the spreader hopper body shall be determined either by calculation or filling with a known volume of material to determine conformance to 3.8.1. Body measurements for inside width and length shall be taken to verify dimensions specified in 3.8.1.

4.4.2 Lifting and tiedown attachment test. When lifting and tiedown attachments are required, they shall be tested to determine conformance to 3.11.

4.4.3 Operation test. The spreader shall be operated for a sufficient period of time to determine that the equipment is in good operating condition and the spreader meets the requirements of this specification.

4.4.4 Performance test. The spreader shall be mounted as required for type I or type II. With the truck being driven on the highway, the spreader shall be capable of spreading salt or abrasive material evenly over a highway 40 feet (12 192 mm) in width, and shall be so constructed that it is easily adjustable to spreading material in an evenly distributed pattern from 6 feet to 40 feet (1 829 mm to 12 192 mm) in width.

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The spreader covered by this specification is intended for spreading salt, aggregates, and other materials on highways, streets, runways, or construction sites.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of spreader required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2 and 4.2.1).

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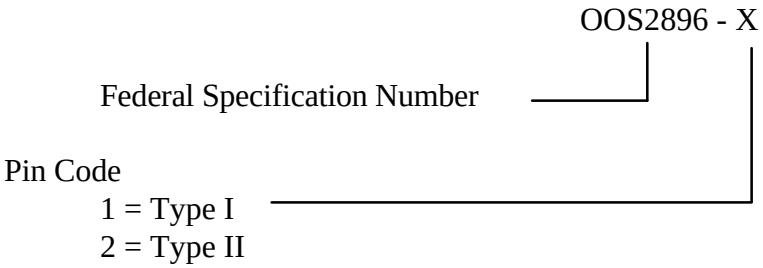
d. Lubrication if different than required (see 3.9).
e. When lifting and tiedown attachments are required (see 3.11).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Classification cross reference. Classifications used in this specification (see 1.2) are identical to those found in superseded Military Specification MIL-S-29208B(YD).

6.5 Part or Identifying Numbers (PINs). The specification number and type are combined to form PINs for material spreaders covered by this document (see 1.2). PINs for material spreaders are established as follows:



6.6 Supersession data. The specification replaces Military Specification MIL-S-29208B(YD) dated 17 January 1992.

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to superseded document.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Navy - YD1

(Project 3895-0038)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** OO-S-2896	**2. DOCUMENT DATE (YYMMDD)** 970314

3. DOCUMENT TITLE
SPREADER, MATERIAL, POWER DRIVEN, TRUCK MOUNTED, COMMERCIAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME RUDY PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial: 805-982-5843 (2) AUTOVON: 551-5843
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC CODE 15E2D 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703)756-2340 AUTOVON 289-2340

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